- Last time:
  - yields disagreed with previous results after generator setup change
  - Poured over MC setup with fine toothed comb, believe numbers are now reliable
  - Generator results after matching and ISR change now agree with previous results in same phase-space
  - The high yields last time were the problem
    - the fiducial volume change was applied inconsistently, inflating our efficiency and resulting in erroneously high yields
    - Now fixed and numbers approx. match previous iterations

- **Conclusion remains the same:**
  - We can claim sensitivity to new and *unique* phase-space
  - Existing searches can't reach it (their backgrounds are too high)
    - This includes the entirely in-time events
  - Margins have changed but not the conclusion

- If we want to increase margin of sensitivity, we have options:
  - combined run 2+3 reinterpretation
    - I think we should do this anyway: CTP applies to both runs, so why wouldn't we?
      - doing so gives ~11 events; need to re-check background
      - Can vary RoI requirement
    - We can vary the MET cut, makes a big difference
      - Would mean re-evaluating background (but suppression factor remains same)

- Next steps: decide on the above -> B2F reco efficiencies (depends on MET choice) and firm in backgrounds -> paper strategy
- Lower yields disappointing, but overall good news

B2F yields (MET > 200 GeV, same cut as MonoJet)

id	mS [GeV]	m_1 [GeV]	$\beta(\chi_1)$ (Scalar rest frame)	Δm_{10}	σ [fb]	N Run2	N Run2+3
100004	450	215.5	0.25	60	32.78	1.57	3.38
100008	450	204.1	0.40	60	74.47	1.53	3.28
100006	450	212.4	0.30	60	46.63	1.29	2.77
100016	600	283.9	0.30	60	8.33	1.15	2.47
100009	450	204.1	0.40	100	71.43	0.98	2.11
100007	450	212.4	0.30	100	44.91	0.92	1.97
100018	600	272.9	0.40	60	14.49	0.90	1.94
100014	600	288.1	0.25	60	5.81	0.76	1.63
100005	450	215.5	0.25	100	31.51	0.65	1.39
100015	600	288.1	0.25	100	5.58	0.61	1.32
100002	450	218.1	0.20	60	21.13	0.58	1.25
100012	600	291.5	0.20	60	3.83	0.58	1.24
100017	600	283.9	0.30	100	8.11	0.45	0.96
100019	600	272.9	0.40	100	14.02	0.38	0.82
100013	600	291.5	0.20	100	3.71	0.35	0.76
100003	450	218.1	0.20	100	20.50	0.28	0.61

B2F yields vs MET threshold

MET trigger efficiency ~90% at 150 GeV, so could go this low if we want, but of course need to then re-estimate background

id	Run 2 (MET > 200)	Run 2+3 (MET > 200)	Run 2 (MET > 175)	Run 2+3 (MET > 175)	Run 2 (MET > 150)	Run 2+3 (MET > 150)
100006	1.29	2.77	3.09	6.65	5.35	11.51
100004	1.57	3.38	2.22	4.77	3.35	7.20
100008	1.53	3.28	2.00	4.29	3.25	6.98
100009	0.98	2.11	1.45	3.12	3.09	6.65
100005	0.65	1.39	0.85	1.82	1.89	4.06
100002	0.58	1.25	1.13	2.44	1.49	3.21
100007	0.92	1.97	1.21	2.61	1.49	3.20
100016	1.15	2.47	1.25	2.70	1.39	3.00
100003	0.28	0.61	0.55	1.18	1.39	2.98
100018	0.90	1.94	0.99	2.14	1.34	2.88
100017	0.45	0.96	0.82	1.76	1.05	2.26
100012	0.58	1.24	0.80	1.72	1.04	2.24
100015	0.61	1.32	0.79	1.71	1.04	2.24
100014	0.76	1.63	0.84	1.80	1.00	2.15
100019	0.38	0.82	0.65	1.40	0.82	1.77

DV+MET

id	B2F R2	their R2	B	S/B	S/sqrtB
100004	1.57	0.872	0.8 ± 0.5	1.090	0.975
100008	1.53	0.595	0.8 ± 0.5	0.744	0.665
100006	1.29	1.154	0.8 ± 0.5	1.443	1.290
100016	1.15	0.368	0.8 ± 0.5	0.460	0.411
100009	0.98	0.768	0.8 ± 0.5	0.960	0.859
100007	0.92	0.473	0.8 ± 0.5	0.591	0.529
100018	0.90	0.710	0.8 ± 0.5	0.888	0.794
100014	0.76	0.249	0.8 ± 0.5	0.311	0.278
100005	0.65	0.053	0.8 ± 0.5	0.066	0.059
100015	0.61	0.203	0.8 ± 0.5	0.254	0.227

MS+1Vtx

id	B2F R2	their R2	B	S/B	S/sqrtB
100004	1.57	0.921	46 ± 9	2.00e-2	0.136
100008	1.53	2.246	46 ± 9	4.88e-2	0.331
100006	1.29	1.237	46 ± 9	2.69e-2	0.182
100016	1.15	0.242	46 ± 9	5.26e-3	0.036
100009	0.98	2.176	46 ± 9	4.73e-2	0.321
100007	0.92	1.372	46 ± 9	2.98e-2	0.202
100018	0.90	0.420	46 ± 9	9.13e-3	0.062
100014	0.76	0.168	46 ± 9	3.65e-3	0.025
100005	0.65	1.005	46 ± 9	2.19e-2	0.148
100015	0.61	0.159	46 ± 9	3.46e-3	0.023